

Yam (*Dioscorea* spp.) cultivation and landraces with market potential in South America

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3.1 Introduction

The disparity in terms of production, harvested area, and yield in yam (*Dioscorea* spp.) cultivation among and between continents and countries is clearly observable in [FAOSTAT \(2021\)](#) data. World annual production in 2019 was estimated at 74 million tons. In South America, the continent produced 704,363 tons in 63,656 ha ([FAOSTAT, 2021](#)). The crop constitutes the predominant starch source in sub-Saharan Africa, where food security for a growing human population is a critical issue. About 93% of total yam production in the world in 2008 was produced in five West African countries (Nigeria, Cote d'Ivoire, Ghana, Benin, and Togo) located in the traditional “Yam Zone” ([Fu et al., 2011](#)). Nigeria is the largest producer, accounting for 70%–76% of world production ([FAOSTAT, 2021](#)). In South America, there are only data from four countries, with Colombia exhibiting the highest production, followed by Brazil, Venezuela, and Guyana ([FAOSTAT, 2021](#)).

A considerable number of cultivated plants introduced to Brazil during the colonial period have been adapted to different cropping systems, with yams being an excellent example, having been adopted by many traditional communities. Brazil's cultural diversity has contributed to the widespread production of different species of yams, but they

remain unknown or underappreciated in many areas (Siqueira, 2011). The literature on yam cultivation published in recent decades lacks adequate information and needs to be revised and organized in order to better identify gaps and research priorities. In South America, and Brazil particularly, current studies and investments in yam research are incipient. Although yam is a well-known crop in parts of Colombia, Brazil, and other countries of South America, it is still considered an “orphan” and underutilized crop (Siqueira & Veasey, 2009), mostly planted by small holders, often using traditional agricultural methods (Nascimento et al., 2015; Silva et al., 2017; Siqueira et al., 2014). They still lack priority in plant breeding projects, as very few institutions in Brazil, if any, are including yams in their projects.

South America’s yam-growing countries have unique and complex histories that involve different colonization processes across a vast territory, leading to the adaptation of agricultural systems to local and temporal conditions, as well as the attribution of diverse common names to cultivated yams. While South America’s regional and cultural diversity have undoubtedly contributed to its agrobiodiversity, yam production is highly localized and maintained at a small scale, with the consequence that yams remain a little-known vegetable crop in large urban centers and among new generations.

3.2 Yam production in South America

Yams (*Dioscorea* spp.) are a staple food for more than 100 million people worldwide and collectively constitute an important crop in South America. Together with other crops, yams are often associated with traditional agricultural systems, being a subsistence food source for many agricultural communities (Lebot, 2009). Despite the important contributions to human health from the active compounds found in several species, yams continue to be marginalized and are still considered to be “orphan” crops, meaning they are underutilized, somewhat forgotten by society and not considered a priority in political agendas (Siqueira & Veasey, 2009; Siqueira, 2011).

Yams were introduced to South America from Africa and Asia during the colonial period and are currently grown in areas with tropical and subtropical climates, primarily in the northern part of the continent. Among the cultivated species, only *Dioscorea trifida* L.f., is native to the region. Despite the global importance of *Dioscorea* spp. as sources of food and pharmaceutical agents, there are few studies describing how and where the edible species were domesticated, and the data are especially poor for *D. trifida*. The *Dioscorea* species that are commercially cultivated for consumption in South America are *Dioscorea alata* L., *Dioscorea trifida*, and *Dioscorea cayenensis* Lam. (the latter referred to in earlier literature as *Dioscorea rotundata* Poir. or the *D. cayenensis*/*D. rotundata* species complex). Brazilian yam production, according to Mesquita (2001), Veasey et al. (2010), Siqueira (2011), and Santos and Passos (2020), is maintained basically in the Northeastern and Southeastern regions, grown mainly by small-scale farmers and/or on small plots, in the states of Paraíba, Pernambuco, Bahia, Alagoas, Minas Gerais, São Paulo, Santa Catarina, and Espírito Santo. In the North of Brazil, the main species grown and commercialized by small-scale farmers is *D. trifida* (Beyerlein & Pereira, 2018; Castro et al., 2012). In the

following paragraphs, we will highlight the main *Dioscorea* spp. observed in Brazil and Colombia, listed according to their cultural and economic importance.

D. alata, known as water yam, is one of the main commercial species in several states of Brazil, presenting edible tubers of high nutritional quality (Soibam et al., 2017). Some of its varieties derive from the “Florida” cultivar, originally from Puerto Rico and introduced to Brazil in the 1950s by the Agronomic Institute in Campinas (IAC), São Paulo state (Siqueira et al., 2012). According to Monteiro and Peressin (2002), this variety, due to its uniform tubers and easy management, has satisfied commercial requirements throughout the years. However, its culinary qualities are not as appreciated as the “Mimoso” variety, which was extensively grown in the past but is difficult to find nowadays in the market, mainly due to its susceptibility to anthracnose (*Colletotrichum gloeosporioides* Penz.). Siqueira et al. (2014) provided an overview of the cultivation and distribution of *D. alata* in four Brazilian regions (South, Southeast, Northeast, and Central West). The authors observed a wide distribution of this species with a variety of vernacular names. Most farmers cultivate yams in fields; however, home gardens were also found to be an important site for the cultivation of this tuber. Additionally, most farmers cultivate yams in association with other crops in areas of different sizes.

The Northeast region of Brazil accounts for about 90% of national production of *Dioscorea* spp., especially the states of Pernambuco, Paraíba, Alagoas, Bahia, and Maranhão (Silva et al., 2017). In these states, the predominant species is the Guinea yam, also referred to as the African yam complex (*D. cayenensis*/*D. rotundata*), followed by *D. alata* (Silva et al., 2017; Veasey et al., 2010). *Dioscorea cayenensis*/*D. rotundata* is the species with the highest rate of tuber production in the world, while *D. alata* has the greatest number of varieties over the broadest geographic distribution (Montaldo, 1991). The species *D. cayenensis* subsp. *cayenensis* has a characteristic yellow tuber, while *D. rotundata* subsp. *rotundata* has a white tuber, although there are a large number of cultivars with an intermediate color (Martin, 1978), which leads some authors to consider this as a species complex (Chair et al., 2005; Dansi et al., 2001, 2013; Mignouna et al., 2002; Girma et al., 2014; Zannou et al., 2009). Other authors have classified or identified them as separate species (Bressan et al., 2014; Silva et al., 2016; Tostain et al., 2007). Since 2007, the convention among yam taxonomists has been to treat them as subspecies of *D. cayenensis*: *D.c.* subsp. *cayenensis* and *D.c.* subsp. *rotundata* (Govaerts et al., 2007; POWO, 2023).

In Brazil, species of the *D. cayenensis*/*D. rotundata* complex are cultivated in most states of the Northeast, mainly in the states of Pernambuco, Paraíba, Alagoas, Bahia, and Maranhão, producing cylindrical tubers of variable size, usually from 1 to 10 kg. Their propagation, as well as that of other species of yam, is vegetative, from the tubers, via vegetative propagation (Santos, 1996; Veasey et al., 2010). Besides their high number of food uses,

D. cayenensis/rotundata are also cultivated for their medicinal properties, in which steroidal saponinins stand out, used in the production of cortisone and synthetic hormones (Avula et al., 2014).

Dioscorea trifida occurs in areas of rain forests, savannas of the Planalto Central, rocky fields of the *Cadeia do Espinhaço*, and seasonal forests adjacent to these fields (Pedralli et al., 2002). This species is cultivated by small-scale farmers in the State of Amazonas (Beyerlein & Pereira, 2018; Castro et al., 2012) and also maintained and propagated by

traditional farmers in the *Vale do Ribeira* and *Ubatuba*, São Paulo, in *Baixada Cuiabana*, Mato Grosso, and in *Joinville* and *São Francisco do Sul*, Santa Catarina (Bressan et al., 2005; Ferreira et al., 2010; Nascimento et al., 2013, 2015). Elsewhere in South America, it is cultivated by indigenous groups throughout the Amazon, Guyana Shield, and Chocó biogeographic regions, as well as in the Sierra Nevada de Santa Marta (Raz, 2015). Despite the geocultural and socioeconomic importance of *D. trifida*, few studies have been conducted to explore its potential and to develop conservation strategies for this crop (Beyerlein & Pereira, 2018; Beyerlein et al., 2019; Nascimento et al., 2015).

As the cultivation and consumption of yam are very intense in family agricultural production systems practiced by traditional communities, these systems provide a favorable environment for the generation and maintenance of genetic diversity of this crop (Veasey et al., 2010). Despite the problems related to agricultural production and lack of marketing initiatives, yams are an important food source, considering that they are highly nutritious (Soibam et al., 2017), and have astringent, antimicrobial, diuretic, and energizing properties, making them an important component in the fight against malnutrition and metabolic disorders such as diabetes and high cholesterol (Ramos-Escudero et al., 2010).

An interesting observation from our studies is that most farmers (62%) plant more than one variety of yams of the same species, or more than one *Dioscorea* species in their lands, which according to the farmers, increases the probability of obtaining sufficient production to guarantee the harvest of the crop and their families' sustenance (Veasey et al., 2010). Beyerlein (2017) also observed, in study sites in Amazonas, Brazil, that farmers cultivate up to five varieties, with most of them cultivating a commercial variety and keeping one to four other varieties in smaller quantities for their own consumption or for sale by special order. A related observation is that farmers who maintain more than one variety have mostly grown yams from childhood. Hence, the maintenance of traditional agricultural systems is important for preserving agrobiodiversity and associated knowledge.

3.2.1 Regions of yam cultivation

The total cultivated area for yam production in the world increased, on average, from 2.06 million ha in the 1970s to 8.83 million ha in 2020. Africa is the continent with the largest area, 8.69 million ha, followed by America (101,798 ha), Oceania (28,974 ha), and Asia (9,899 ha) (FAOSTAT, 2021).

Among the African countries, Nigeria stands out, with 6.30 million ha, which represents around 71% of the world's cultivated area for yam production. Next, we find Côte d'Ivoire (1.20 million ha), Ghana (468,433 ha), Benin (228,998 ha), and Togo (98,547 ha), which together with Nigeria account for around 96% of the world's total yam cultivation area (FAOSTAT, 2021).

On the American continent, South America (64,558 ha) has the largest cultivation area, followed by Caribbean countries (32,796) and Central America (4,444 ha). North America is not represented in the databases because it has no significant yam growing areas. Considering only the Caribbean countries, Haiti is the main country in terms of yam cultivation, with 9983 ha, which represents around 30% of the yam cultivated area in this macroregion. Haiti, together with Jamaica, Cuba, and the Dominican Republic, accounting

for 31,185 ha, which corresponds around 95% of the yam planted area in Caribbean America. In Central America, Panama is the country with the largest planted area, with 3,426 ha (FAOSTAT, 2021).

In South America, the continent richest in *Dioscorea* species (Govaerts et al., 2007), the area under yam cultivation has fluctuated substantially in recent years, but the overall tendency has been an increase during the period from 1970 to 2019. The countries with the largest yam cultivated areas are Colombia (34,508 ha) and Brazil (25,428 ha), followed by Venezuela (4,601 ha). Brazil is the 12th country in yam planted area in the world (FAOSTAT, 2021).

In Colombia, the country with the largest cultivated area in the Americas and with 52 native species (Raz, 2015), there is little documentation on the use of *Dioscorea* spp., with the exception of the two most economically important species (*D. alata* and *D. cayenensis* subsp. *rotundata*). These are widely cultivated, although commercial production is centered in the Colombian Caribbean region, where culinary traditions are strongly influenced by the presence of a large population of African descent. The South American native species, *D. trifida*, is also widely cultivated, mainly among indigenous communities located in lowland areas of the country (Pinzón-Rico and Raz, 2017). Colombia produced 424,000 tons of yam in 2020 (FAOSTAT). Commercially the most important species is *D. alata*, but as in Brazil, production has been impacted over the last few decades by anthracnosis (Campo Arana, 2000). The varieties “criollo” and “diamante,” the latter developed in Costa Rica, are the main cultivars of this species. *Dioscorea cayenensis* subsp. *rotundata* is the second most important commercial species in Colombia, known by the varietal name “espino.” Locally, this species has also been investigated as a potential source of starches for the cosmetics industry (Zuluaga et al., 2007).

As in Colombia, in Brazil, the most cultivated species are *D. cayenensis*/*D. rotundata* and *D. alata*, followed by *D. trifida*. These are mainly cultivated in the Northeast and Southeast regions (Ferreira et al., 2010; Santos, 1996; Veasey et al., 2010), and in the North region in the case of *D. trifida* (Beyerlein & Pereira, 2018; Castro et al., 2012). Unlike Colombia and Brazil, the other South American countries do not have a significant cultivated area.

3.2.2 Farming systems

In many parts of South America, yams are grown in multicrop systems rather than in monocultures, regardless of whether they are grown for sale or personal consumption. In Brazil, some states use yam monoculture, but this cultivation format is uncommon (Fig. 3.1A and B). In the Caribbean Region of Colombia, yams are typically integrated with corn or cassava and secondary crops such as papaya, watermelon, capsicum peppers, or even woody legumes such as the pigeon pea (Alvarez Soto, 2000). Multicrop systems have the advantage that they do not require stakes to support the growing yam vines. In contrast, Afro-Caribbean communities in Jamaica and other parts of the Antilles generally grow yams on stakes (yam sticks) in monocultures (Fig. 3.1C and D). Yams must be grown in well-drained soils and are typically propagated vegetatively using whole tubers or portions of tubers weighing between 100–150 g (Alvarez Soto, 2000). Seed tubers are usually planted to a depth of about 10 cm and weeds are managed using chemical or manual control methods. Local agronomists often recommend the application of fertilizers and insecticides, but these are not always available to farmers. Pathogens are extremely



FIGURE 3.1 Yam monoculture in São Paulo State, Brazil (A and B) and yam plantation in Jamaica (C and D). Photographs C and D: courtesy of Vilpoux, O.F.

difficult to control in the field, but in vitro methods have been used successfully in Colombia to propagate vegetative material free from viral and fungal pathogens (Perea Dallas & Buitrago Hurtado, 2000).

Today in Colombia, about 30,000 families participate in commercial yam cultivation and most are smallholders who often rent, rather than own land. Within the last decade, they have organized at the national level a united federation of yam growers, *Federación Nacional de Productores de Ñame* (Fedename, 2022), which has helped to secure better prices and access to markets. In 2019, the Colombian Ministry of Agriculture also consolidated a National Council to oversee the yam supply chain and support agrobusinesses, the first such organization to exist in South America (*Consejo Nacional de la Cadena del Ñame y su Agroindustria*).

Many of South America's yam growing regions are located near the Caribbean or Atlantic Coasts and are inhabited by Afro descendent or mestizo populations, with the prominent exception of the Amazon region, which is the likely origin of *D. trifida* (Bousalem et al., 2010). Today many indigenous communities in the Amazon continue to cultivate this species using traditional methods without chemical inputs. There is a wealth

of ethnographic and ethnobotanical literature describing practices associated with swidden agriculture in different parts of the Amazon Basin (Deneven, 2001). These are multicrop systems (*chagra* or *chakra* in Spanish) in which cassava is generally the dominant food crop and *D. trifida* is a secondary component, often together with other monocotyledonous species with edible rhizomes (*Heliconia hirsuta*, *Maranta arundinacea*, *Xanthosoma* spp., etc.). Occasionally, *D. alata*, *D. cayenensis*, and even *Dioscorea bulbifera* may also be found in Amazonian swidden fallows (personal observation). Indigenous farmers, like their counterparts in rural *campesino* communities, also do not use stakes to support growing yam vines. Crops produced in these systems are generally for local consumption, although communities living near urban areas may sell a portion of their yield to supplement their incomes (Sierra & Raz, 2014).

3.2.3 Distinguishing features of the cultivated yams

The three main cultivated *Dioscorea* species in South America can be distinguished by differences in both the aerial plant parts and the tubers. *D. cayenensis/rotundata* have spiny, cylindric stems and cordate leaves that can be opposite or alternate and are usually somewhat leathery and shiny on the upper surface. *Dioscorea alata* has angular stems with membranaceous wings that run along the entire stem length. The leaves are opposite with cordate to sagittate bases and small pseudostipules at the bases of the petioles; the blades have a papery texture. Aerial bulbils of irregular shape are sometimes produced in the leaf axils of this species, but these are not harvested for human consumption. *Dioscorea alata* varieties in this region tend to be white fleshed, although some purple-fleshed varieties popular in Asia may be of future commercial interest. Tuber size and shape are not reliable characteristics to differentiate *D. cayenensis/rotundata* from *D. alata*, but *D. cayenensis* generally has drier, more fibrous tubers, while those of *D. alata* tends to have less fiber and more mucilage when cut. Both species exhibit variation in flesh color, with *D. cayenensis* subsp. *cayenensis* generally having



FIGURE 3.2 *Dioscorea cayenensis/rotundata* in a Brazilian northeast coastal cultivation system and a public market showing the white flesh color tubers. Source: Photograph courtesy of Paulo César Tavares de Melo.



FIGURE 3.3 White and purple flesh color in *Dioscorea trifida* collected in Santa Catarina, Brazil.

yellow fleshed tubers and *D. cayenensis* subsp. *rotundata* generally having white fleshed tubers. White fleshed varieties are much more common in South America (Fig. 3.2).

Dioscorea trifida is the easiest to distinguish from this group, morphologically. It has winged stems and petioles, often with purple pigmentation, and alternate leaves with 3–7 lobes and a papery texture. A single plant produces numerous small tubers (usually less than 15 cm long and 7 cm in diameter) that are covered by a thinner, smoother skin than the two other species, which have a courser epidermis that is darker brown in color and more rough textured. The variation in flesh color in *D. trifida* is along the spectrum of white to purple with many intermediaries (Fig. 3.3). The tubers of *D. trifida* are sweeter than those of the other species and have a smoother, less grainy mouth feel. Individual tubers of *D. cayenensis* subsp. *rotundata* can grow up to more than 100 kg, and in Colombia, the largest of these are exhibited by local farmers at the annual National Yam Festival in San Cayetano, near the Caribbean coast.

D. bulbifera L. is occasionally grown by farmers in Brazil, Colombia, and elsewhere in South America, but it is the least culturally and economically important of the species included here. It has cylindrical stems without spines or wings, alternate leaves that are broadly cordate and with conspicuous pseudostipules at the base of the petioles (Fig. 3.4). It produces more or less round bulbils (air potatoes) in the leaf axils, and these can be eaten, although some varieties are toxic. The underground tuber is much less developed than in the other cultivated species and is not harvested for food. It is an aggressively invasive species in the Southeastern United States (Coursey, 1967; Knuth, 1924; Raz, 2002).

3.3 Vernacular names and genetic erosion of local varieties

Dioscorea species, when grown in traditional agricultural systems or on small farms, present wide intraspecific morphological and genetic variation (Beyerlein & Pereira, 2018;



FIGURE 3.4 Inflorescence and aerial tubers of *Dioscorea bulbifera*.

Bressan et al., 2005, 2011, 2014; Castro et al., 2012; Girma et al., 2014; Hasan et al., 2008; Siqueira et al., 2014), which may lead to a diverse folk nomenclature (Table 3.1).

This diversity is due not only to morphological variation but also to cultural variation, considering that different names are given according to their geographic location (different local communities, regions, and countries, as well as cultural values) (Amorozo, 2008; Pedralli et al., 2002). The common name “cará,” of a Tupi-Guarani language origin, has been attributed to yam species from the South, Southeast, Central West, and North regions in Brazil, whereas the common name “inhame” was given to *Dioscorea* plants mainly from the Northeast region (Bressan et al., 2005; Silva et al., 2017; Siqueira, Nascimento, et al., 2014) (Table 3.1). The criteria used to name the different varieties (Amorozo 2008; Emperaire & Peroni 2007) can be based on morphological aspects of the varieties they are describing, such as *cará roxo* (purple yam), *cará branco* (white yam), *cará mandioca* (manioc yam), and *inhame da folha larga* (wide leaf yam); or based on analogies between plants and animals, such as the names *cará pé de elefante* (elephant’s foot yam), *cará mão de anta* (tapir’s hand yam); or based on the name of a person, which could be a farmer who first appeared with it, such as *cará Miguel* (Miguel’s yam), *cará do Joaquim* (Joaquim’s yam); or based on a saint’s day, which may coincide with the harvest time for that variety, such as *cará São João* (Saint John yam) and *inhame São Tomé* (Saint Tomé yam).

Different local names are given by farmers to landraces of a *Dioscorea* species according to the geographic location, such as the names attributed to *D. alata* (Table 3.1). Farmers from the Southeast of Brazil name their local varieties as *cará são joão* (são joão yam), *cará guaçu* (guaçu yam), and *cará mimoso* (cute yam), while those from the Central West name their varieties as *cará lavanca* (lavanca yam), *cará cipó* (vine yam), and *cará manchado* (spotted yam), and those from the Northeast of Brazil name them as *inhame São Tomé* (Saint Tomé yam), *inhame de Pernambuco* (Pernambuco yam), and *inhame roxo bolota* (purple ball yam). Some of the names are repeated in different regions, such as *cará* (yam), *cará roxo* (purple yam), and *cará branco* (white yam) for *D. trifida*; also such as *cará de espinho* (thorn yam) for *D. cayenensis* subsp. *rotundata*, in the South and Southeast regions, given the fact that this species presents lots of thorns on its stems; or *cará moela* (gizzard yam) in the

TABLE 3.1 Local vernacular names given to *Dioscorea* species in different countries of South America.

Species	Country	State/region	Vernacular names	Reference
<i>Dioscorea trifida</i>	Brazil	São Paulo/Southeast	<i>cará São João; cará roxo; cará roxo mirim; cará mirim; cará roxo Indaiá; cará São João roxo; cará São João branco; cará branco; cará peludinho; cará pipa; cará prato; cará una; cará cobrinha</i>	Peroni and Hanazaki (2002) , Bressan et al. (2005) , Nascimento et al. (2015)
		Santa Catarina/South	<i>Cará; cará mimoso; cará pão; cará carcanhá de nego</i>	Nascimento et al. (2015)
		Mato Grosso/Central West	<i>cará roxo; cará branco; cará pé de anta; cará mão de anta; cará do Joaquim; pumbinho branco; cará roxo comprido</i>	Ferreira et al. (2010) , Nascimento et al. (2015)
		Amazonas/Amazon	<i>cará roxo comum; cará branco; cará roxão; cará durão; cará macaxeira; cará pata de onça; cará ovo de cavalo; cará rabo de mucura; cará miguel; cará inhame; cará japones; cará branco comprido; cará boliviano; cará barbudo; cará casado; cará pé de porco; cará pé de burro</i> <i>cará preto; cará roxo de humaitá; cará mão de onça</i> <i>cará pé de elefante; cará branco redondo; branco casquinha roxa; cará açai redondo; cará açai comprido</i>	Castro et al. (2012) , Beyerlein and Pereira (2018)
		Alagoas and Bahia/Northeast	<i>inhambu; inhame da mata</i>	Silva et al. (2020) , Medeiros et al. (2021)
		Venezuela	Amazonas State	<i>mapuey blanco; mapuey morado rugoso; mapuey morado liso</i>
	Peru	District of Rupa-Rupa	<i>sachapapa purple tuber</i>	Ramos-Escudero et al. (2010)
	Colombia	Orinoco region	<i>tabena</i>	Bernal et al. (2017)
		Caribbean region	<i>ñampín</i>	Alvarez Soto (2000)
	<i>Dioscorea alata</i>	Brazil	São Paulo/Southeast	<i>cará guaçu; cará guaçu médio; cará aipim; cará branco; cará roxo; cará são João; cará são João roxo; cará são João branco; cará; cará bolo; cará coco; cará pé de cavalo; cará Flórida; cará mimoso</i>
Mato Grosso/Central West			<i>cará branco; cará roxo; cará cará lavanca; cará arroba; cará pele roxa; cará canga; cará cipó; cará manchado; cará mandioca; cará inhame</i>	Ferreira et al. (2010) , Siqueira et al. (2014)
Pernambuco, Paraíba, Piauí, Maranhão/Northeast			<i>inhame São Tomé; inhame branco; inhame roxo; inhame roxo bolota; inhame babão; inhame mandioca; inhame de pernambuco</i>	Siqueira et al. (2014)
Santa Catarina/South			<i>cará roxo; cará; cará pão; cará pão roxo; cará pão branco</i>	Siqueira et al. (2014)
Colombia, Caribbean Region			<i>Diamante; ñame criollo; ñame plateño; ñame baboso</i>	Bernal et al. (2017)

<i>Dioscorea cayenensis</i> and/or <i>Dioscorea rotundata</i>	Brazil	São Paulo/Southeast	<i>cará de espinho; cará d'angola; cará guaçu; cará São João roxo; cará japeira; cará espinhento</i>	Peroni and Hanazaki (2002), Bressan et al. (2014), Silva et al. (2017)
		Paraíba and Pernambuco/Northeast	<i>inhame da Costa; inhame de Pernambuco; inhame japecanga; inhame branco; inhame; inhame original; inhame da folha larga</i>	Silva et al. (2017)
		Santa Catarina/South	<i>cará de espinho</i>	Silva et al. (2017)
	Colombia, Caribbean Region		<i>ñame espino; ñame mejorado</i>	
<i>Dioscorea bulbifera</i>	Brazil	São Paulo and Minas Gerais/Southeast	<i>cará frissura; cará moela; cará do ar; cará paquera; cará de árvore; cará de cipó; cará fígado; cará preto</i>	Peroni and Hanazaki (2002); Bressan et al. (2005), Silva et al. (2016)
		Goiás and Mato Grosso/Central West	<i>cará moela</i>	Silva et al. (2016)
		Piauí and Pernambuco/Northeast	<i>inhame fígado; cará moela</i>	Silva et al. (2016)
	Colombia, Caribbean Region		<i>papa de aire; ñame volador; ñame papa</i>	Bernal et al. (2017)

three regions where *D. bulbifera* was found, mainly due to the shape of the aerial tubers of this species, which resemble a gizzard. These are all names related to the morphology of the species. Little was found in the literature of landraces and their local names in other countries of South America. The names *mapuey blanco*, *mapuey morado rugoso*, and *mapuey morado liso* are given to *D. trifida* in Venezuela (Perez et al., 2011). And the name *sacha papa purple yam* was found in papers from Peru (Alva et al., 2017; Ramos-Escudero et al., 2010). In Colombia, the common name *tabena* is applied to *D. trifida* in the northeastern part of the country where that species was commercially cultivated during the 19th century. Further historical research is needed to understand the factors that led to the demise of this market. Given the >300 extant indigenous languages in the Amazon (van der Voort et al., 2021), there may well be hundreds of names for *D. trifida* in this region and other lowland areas of South America where this species is traditionally cultivated, but it is beyond the scope of this review to compile them. The word *ñame* in Spanish, is applied almost universally in Colombia and Venezuela to the *Dioscorea* species originating from Africa and Asia, and as is the case with *cará* in Brazil, additional modifiers are appended to specify local varieties (Table 3.1).

Farmers who practice traditional agriculture have been acknowledged for their roles in creating and/or maintaining high levels of diversity at the local level. Some well documented examples include cassava (Alves-Pereira et al., 2011; Amorozo, 2008; Carrasco et al., 2016; Emperaire & Peroni, 2007; Marchetti et al., 2013; Perez et al., 2019; Siqueira et al., 2010) and yams (Beyerlein & Pereira, 2018; Bressan et al., 2011; Castañeda-Cardona et al., 2020; Chair et al., 2010; Loko et al., 2016; Nascimento et al., 2013; Silva et al., 2016, 2017; Siqueira et al., 2014). However, it is also quite common in traditional or small-scale agriculture to have farmers stating that they used to plant more varieties of a certain species than at the present date (Bressan et al., 2005; Castro et al., 2012; Marchetti et al., 2013; Nascimento et al., 2015; Silva et al., 2017), which may be an indication of genetic erosion occurring in these study regions.

Genetic erosion, defined as the loss of variability from crop populations following social, economical, and agricultural changes (Brush, 1999), can occur at three levels: (1) the crop, (2) the variety, and (3) the allele (Van de Wouw et al., 2009). Many factors may be responsible for it to occur, including urbanization and changes in food preferences; the adoption of modern agricultural practices (mechanization, irrigation, and crop specialization); climate change, natural hazards, and environmental degradation; or due to conflicts and population displacements (Delêtre et al., 2012; Van de Wouw et al., 2009). In a field collection of sweet potatoes (*Ipomoea batatas*) and yams (*Dioscorea* spp.) conducted in Vale do Ribeira, São Paulo State, farmers reported that they cultivated varieties in the past that they no longer planted, and the reasons were lack of secure land tenure, environmental laws restricting farmer's ability to open new areas for farming, rural exodus, and the shortage of family labor (Bressan et al., 2005). Even so, the authors found high diversity in Vale do Ribeira, where farmers maintained four *Dioscorea* species, namely, *D. trifida* (36% of the acquired varieties were from this species), *D. cayenensis* (26%), *D. alata* (22%), and *D. bulbifera* (16%).

Further studies with yam species in Brazil, including a wider sampling in the South, South-East, Central-West, North, and Northeast regions, also reported the existence of local yam varieties that are no longer planted in the farmer's property (Castro et al., 2012;

Nascimento et al., 2013; Pedralli, 1990; Silva et al., 2017). For *D. trifida*, 96% of the interviewed farmers provided such a statement (Nascimento et al., 2013). In the municipality of Caapiranga, in the State of Amazonas, North of Brazil, where *D. trifida* is the most important commercialized yam species, Castro et al. (2012) also mentioned four local varieties of yams that are no longer cultivated by the farmers, and at least nine local varieties that are well known by many but planted by only a few farmers. The same has been observed for cassava. Emperaire et al. (2016) observed that the number of local varieties of cassava planted in Rio Negro, in the Amazon, was reduced by half in 10 years. Marchetti et al. (2013) compared the cassava diversity in 1992 with that found in 2011 in a traditional agriculture area in the State of Mato Grosso. The authors found similar diversity but noticed that the number of varieties per farmer was greatly reduced (9.4 vs 5.4) and reported a decrease in agricultural activity, including a decrease in the size of cultivated area. The authors also noticed that most young people were not engaged in agricultural activities.

The main reasons listed by the farmers for reducing the number of varieties previously planted were abandoning the fields to obtain employment in urban areas; changes in the physicochemical conditions of the soil, leading to the emergence of diseases; reduced labor; unequal land distribution, reducing the area available for agricultural activities; lack of information regarding the proper management of the crop; as well as increased production costs, low returns and lack of financial incentives to encourage the practice of family agriculture (Nascimento et al., 2013; Silva et al., 2017). Similar reasons have been reported by yam farmers in Africa, including the occurrence of pests and diseases, poor soils, drought, flooding, or high soil moisture, causing severe yield losses and genetic erosion (Dansi et al., 2013). Another important reason is related to the preferences of farmers for the higher commercial value of certain varieties, and the influence of eating habits, leading to the preference of certain varieties due to their flavors and colors (Castro et al., 2012). In the case of cassava, Siviero and Lessa (2020) reported that the maintenance of particular varieties in the field is determined by market pressures, specifically demand for products such as water flour, dried flour, *tapioca*, gum, *tucupi*, *maniçoba*, *macaxeira* (sweet manioc), and others.

3.4 Market prospects and culinary uses

Opportunities in the yam market have been increasing in recent years. However, issues related to postharvest practices, yam deterioration, storage, and consumer demand remain neglected areas of research (Iddi et al., 2018). There are also problems associated with the social organization of farmers, which often puts yam producers and traders in vulnerable situations to deal with market demands (Aidoo et al., 2012). In this scenario, consumers are part of the chain and constitute the driving force of the entire marketing process (Iddi et al., 2018), since their demands, tastes, and preferences are determining factors for the type, quantity, and quality of yam.

The word yam is of African origin (*nham*), meaning “to eat.” In European languages, similar words are attributed to this crop, such as *ñame*, *iñame*, *igname*, *yame*, *yam*, and *yams* (Cascudo, 2011). In the Congo, yam has been used traditionally to make flour and in

Guinea for making bread (Gandavo, 1924). References from 15th and 16th-century explorers mention yam cooked and roasted in Senegal, as well as by Sudanese and Bantu peoples (Cascudo, 2011). It was generally considered the favorite food of West Africans. In some regions of Ghana, yams are offered as wedding gifts to brides and, in the case of traditional ceremonies, are offered to guests of honor (Iddi et al., 2018).

If the yam is not marketed in its natural state, processing can be carried out to generate products such as yam powder, breads, and pies (Iddi et al., 2018). Currently, in Brazil, yams are consumed roasted and more commonly boiled (Cascudo, 2011; Dória, 2008; Querino, 1957). Regardless of the method of preparation, yams have a low sugar content, in addition to a higher percentage of starch and protein (Cascudo, 2011; Freitas et al., 2004). For more information about yam composition, read Chapter 8 of the 1st volume of this series.

In addition to their carbohydrate content, yams also provide a number of desirable nutritional components and health benefits such as antioxidant, hypoglycemic, hypocholesterolemic, antimicrobial, and immunomodulatory activities (Padhan & Panda, 2020).

In the book *History of Alimentation in Brazil*, Cascudo (2011) presents the greatest historical and sociological record on Brazilian cuisine and discusses the use of yam in the country's food. This work surveys Brazilian food traditions, the result of miscegenation between indigenous peoples originating in Brazil, the enslaved African population, and the Portuguese. In Brazil, yams were notably involved in the food practices and customs of the indigenous *Tupiniquins* (Cascudo, 2011), a group originating from the Tupi linguistic trunk, who inhabited areas along the Brazilian coast (Eriksen, 2011). Pero Vaz de Caminha, a naval clerk and the country's first chronicler, considered the yam indispensable, a common food and the most visible and salient delicacy of indigenous meals in Porto Seguro, in the state of Bahia (Cascudo, 2011).

Montanari (2008), in his book *Food as Culture*, defends the importance of cataloging and recording recipes. He used the term "written cuisine", which allows a person to codify in an established repertoire the practices and techniques involved in the preparation of food in different societies, in a certain period and space, following a common and collective pattern within each context. The second culinary and gastronomic work published in Brazil, named *Cozinheiro Nacional* (Dória, 2008), presents five special recipes for Brazilian culinary preparations involving yam: *Cará Refogado* (braised yam), *Cará com Açúcar* (yam with sugar), *Cará Ensopado* (stew yam), *Caratinga Cozida* (cooked caratinga), and *Inhame Cozido* (cooked yam). These recipes consist of real delicacies that are easily combined with all types of meats. In the Medicinal Soups section of this same book, there is the recipe for "Porridge to purify the blood," whose main base is yam milk, accompanied by the following ingredients: watercress, lettuce, sugar, almonds, and nutmeg.

The book *Culinary Art in Bahia* (Querino, 1957) has three typical recipes with yam: yam balls (large balls made with cooked yam dough, washed in lemon before cooking and pressed in the pestle, served with caruru—sautéed okra—or efó—paste obtained from some vegetables, flavored with dried shrimp, chilli pepper, and other condiments), Bobó de Yam (yams cooked in small pieces, put on the fire with water and finally seasoned like efó), and Ipêtê (yams cut into small pieces, boiled until they lose their consistency, seasoned with fragrant oil, shrimp, onion, and pepper, the latter grated on the stone).

The annual *Festival Nacional del Ñame* (National Yam Festival) in the Colombian department of Bolívar celebrates the culture and agrobiodiversity of the *Montes de María*, a former conflict zone in Colombia's Caribbean region. The festival features cooking contests in which local women compete in savory and sweet food categories. Among the most celebrated traditional dishes are the dessert *dulce de ñame*, made with milk and sugar, and the savory *mote de queso*, a soup made from boiled yams and *queso costeño*, a salty homemade cheese that is consumed throughout the region. In recent years, the festival has stimulated the creation of a local women's cooperative which has received small grants to scale up commercial food production.

The versatility of yam's culinary uses ranges from meeting basic food needs, being considered one of the staple crops for food security in many countries, to the production of processed products in a commercial and *haute cuisine* context. The cultivation of local varieties of yam in traditional agricultural systems ensures the maintenance of genetic diversity by farmers, who can in turn offer the market different varieties to meet the different preferences and tastes of consumers.

3.5 Final considerations

Yams are of great cultural importance to diverse communities in South America, but they remain unknown or underappreciated by policymakers and populations in large urban areas. We would like to see greater investment in research from both the public and private sectors to characterize and conserve genetic diversity in cultivated yam species and their wild relatives and to bolster crop improvement programs to promote drought and disease resistance.

Local farmers also require consistent support to increase production, formalize distribution networks, and maintain supply chains. Investments in market research and collaboration with communities on product development should also be encouraged to help broaden the appeal of yams and help put them on menu in many more homes and restaurants across South America and around the world.

Acknowledgment

M.V.B.M.S. thank State University of Minas Gerais for the scholarship granted (process no. 2350.01.0003474/2022-12) and W.F.N. thank the support of the Federal University of Maranhão. We also thank CNPq for the scholarship granted to E.A.V. (process no. 309445/2020-5).

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